



Required Capstone Project: Module 4

Due Jun 17 by 11:59pm**Points** None**Available** after Jun 3 at 12am

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Learning Outcomes Addressed

- Understand how AI, GenAI and Agentic AI work and where they apply in business.

- Use prompts effectively for analysis, decision-making and content generation.
- Map business workflows and identify where AI can drive automation and efficiency.
- Evaluate AI opportunities using feasibility, ROI and impact criteria.
- Build AI adoption roadmaps aligned to business goals and KPIs.
- Apply data governance, privacy and ethical AI principles.

Phase 2 – Efficiency & Disruptive Impact

Milestone 4: Efficiency & Financial Impact Model

Estimated Effort: 2 Hours

Objective

In **Phase 1 of the capstone**, you identified and analysed a business workflow that could benefit from Generative AI.

Specifically:

- In **Module 1**, you selected a business workflow and created an **AI Opportunity Brief**.
- In **Module 2**, you analysed the workflow using a **Workflow Baseline & Risk Canvas**.
- In **Module 3**, you designed **structured prompts** that demonstrate how Generative AI could assist with tasks within that workflow.

In this module, you will estimate the **operational and financial impact of applying Generative AI to that workflow**.

You will identify repetitive tasks within the workflow that could be improved using Generative AI and estimate:

- potential **time savings**
- potential **productivity improvements**
- potential **financial cost reductions**

You will also estimate the **implementation cost of introducing AI tools** and calculate a simplified **3-year return on investment (ROI)**.

This exercise reflects how organisations build **business cases for AI investments**.

Note: This is a required assignment and counts towards programme completion.

Task Overview

Using the workflow you selected earlier in the capstone, identify **five repetitive tasks** that could be improved using Generative AI.

For each task, estimate:

- percentage of time saved
- annual hours saved
- cost per employee hour
- annual cost reduction

Then estimate the **cost of implementing the AI solution** and calculate the **3-year ROI**.

Exact numbers are **not required**. Reasonable estimates are acceptable.

Step 1 — Identify Repetitive Workflow Tasks

Review the workflow you analysed in **Module 2**.

Identify **five tasks** that involve repetitive information processing, such as:

- document review
- data extraction
- report preparation
- message analysis
- summarisation tasks

These tasks are often strong candidates for **Generative AI efficiency improvements**.

Example for Reference

Workflow: Vendor Invoice Processing

Possible repetitive tasks:

Task	Description
Invoice data extraction	Extract invoice details manually
Invoice validation	Check invoice against purchase order

Task	Description
Email summarisation	Summarise vendor email queries
Discrepancy review	Identify mismatched invoice details
Monthly finance reporting	Compile invoice statistics

These tasks will be used in the remaining steps.

Step 2 — Estimate Time Savings

Estimate how long each task currently takes and how much time AI assistance could reduce.

Express the improvement as **percentage of time saved**.

Example for Reference

Task	Current Time	AI-Assisted Time	% Time Saved
Invoice data extraction	10 min	4 min	60%
Invoice validation	8 min	5 min	38%
Email summarisation	6 min	2 min	67%
Discrepancy review	12 min	7 min	42%
Finance reporting	30 min	15 min	50%

These estimates represent how AI reduces the time required for each task.

Step 3 — Calculate Annual Hours Saved

Estimate how frequently each task occurs and calculate the **total annual time saved**.

Formula:

Annual Hours Saved = (Time Saved per Task × Annual Task Volume) ÷ 60

Example for Reference

Task	Time Saved per Task	Tasks per Month	Annual Hours Saved
Invoice data extraction	6 min	1,500	1,800 hrs
Invoice validation	3 min	1,500	900 hrs
Email summarisation	4 min	300	240 hrs

Task	Time Saved per Task	Tasks per Month	Annual Hours Saved
Discrepancy review	5 min	200	200 hrs
Finance reporting	15 min	80	240 hrs

This table shows how small time savings per task can produce **significant annual productivity gains**.

Step 4 — Estimate Cost Per Hour

Estimate the **average employee cost per hour** for the role performing these tasks.

This can include salary, benefits, and overhead costs.

If exact numbers are unavailable, use a **reasonable industry estimate**.

Example for Reference

Finance operations analyst

Estimated hourly cost:

\$40 per hour

This value will be used in the next step to estimate cost savings.

Step 5 — Calculate Annual Cost Reduction

Multiply **annual hours saved** by **cost per hour**.

Formula:

Annual Cost Reduction = Annual Hours Saved × Cost per Hour

Example for Reference

Task	Annual Hours Saved	Cost per Hour	Annual Cost Reduction
Invoice extraction	1,800	\$40	\$72,000
Invoice validation	900	\$40	\$36,000
Email summarisation	240	\$40	\$9,600
Discrepancy review	200	\$40	\$8,000
Finance reporting	240	\$40	\$9,600

Total Estimated Annual Savings:

\$135,200

Step 6 — Estimate Implementation Cost

Next, estimate the cost of implementing the AI solution.

This may include:

- AI software subscription
 - system integration
 - implementation setup
 - staff training
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Example for Reference

Cost Category	Estimated Cost
AI platform subscription	\$15,000 per year
Implementation setup	\$20,000
Staff training	\$5,000

Total Initial Implementation Cost:

\$25,000

Annual tool cost:

\$15,000

Step 7 — Estimate 3-Year ROI

Now estimate the **3-year financial return**.

Step 7A — Calculate Total 3-Year Savings

Annual savings:

\$135,200

3-year savings:

$\$135,200 \times 3 = \$405,600$

Step 7B — Calculate Total Investment

Implementation cost: \$25,000

3 years of AI software:

$$\$15,000 \times 3 = \$45,000$$

Total investment:

\$70,000

Step 7C — Calculate ROI

Formula:

$$\text{ROI} = (\text{Total Savings} - \text{Total Investment}) \div \text{Total Investment}$$

Calculation:

$$(\$405,600 - \$70,000) \div \$70,000$$

ROI:

4.79 or 479% ROI

This shows the potential financial return created through workflow efficiency improvements.

Required Output Structure

Your final submission should include a summary table similar to the following.

Task	% Time Saved	Annual Hours Saved	Cost per Hour	Annual Cost Reduction
Task 1				
Task 2				
Task 3				
Task 4				
Task 5				

Also include:

- estimated **implementation cost**
- estimated **3-year ROI**

Practical Tips

Focus on Repetitive Tasks

Generative AI is most effective when applied to:

- document-heavy workflows
- repetitive reporting
- summarisation tasks
- information extraction

Use Reasonable Estimates

Exact financial data is not required.

Use **realistic approximations** to illustrate the potential business impact.

Keep Calculations Simple

This exercise demonstrates **basic AI business case modelling**, not detailed financial forecasting.

Submission

Before submitting your **Efficiency & Financial Impact Model**, confirm that you have included:

- ✓ Five repetitive workflow tasks
- ✓ Estimated percentage of time saved
- ✓ Estimated annual hours saved

- ✓ Cost per hour assumption
- ✓ Annual cost reduction calculation
- ✓ Estimated implementation cost
- ✓ Estimated 3-year ROI

Instructions:

- Use the same document that you downloaded in Module 1 and include your responses.
- We recommend you add additional text boxes or slides (if needed) for each milestone task to ensure that your responses are documented in a proper format.
- The document will gradually grow as you complete each capstone milestone. The final submission occurs at the end of Module 14.

Grading Rubric

Maximum Points: 20

Criteria	Full Points	0 Points
Identification of Repetitive Workflow Tasks (4 points)	The submission identifies five repetitive tasks within the selected workflow that are suitable for Generative AI-based improvement. (4 points)	Workflow tasks are missing. (0 points)
Time Savings Estimation (4 points)	The submission includes estimates for percentage of time saved and annual hours saved for all identified tasks. Calculations are clearly presented. (4 points)	Time-saving estimates or annual hours calculations are missing. (0 points)
Cost Per Hour & Annual Cost Reduction (4 points)	The submission includes estimated employee cost per hour and annual cost reduction calculations for the identified tasks. Financial estimates are logical and clearly presented. (4 points)	Cost assumptions or annual cost reduction calculations are missing. (0 points)

Criteria	Full Points	0 Points
Implementation Cost Estimation (4 points)	The submission includes estimates of AI implementation costs, such as software, setup, integration, or training costs. (4 points)	Implementation cost estimates are missing. (0 points)
3-Year ROI Estimation (4 points)	The submission includes a calculated 3-year ROI estimate based on projected savings and implementation investment. (4 points)	ROI calculations are missing. (0 points)

Why this Step Matters

Organisations rarely adopt AI based on technical capability alone.

They invest in AI when it demonstrates **clear economic value**.

This exercise helps you learn how to:

- quantify productivity improvements
- estimate financial benefits of AI automation
- build a business case for AI adoption
- communicate AI value to leadership

In the next module, you will explore how Generative AI can move beyond **efficiency improvements** to create **disruptive business impact**.